

Notes: Bonaparte and Kumar (JFE, 2013)

Political Activism, Information Costs, and Stock Market Participation

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1 Big picture

- **Question.** Does political activism make households more likely to participate in the stock market?
- **Core hypothesis.** Politically active people follow political news more closely. Because political news and business news are often related, they face **lower information-gathering costs** for learning about financial markets.
- **Setting.** The paper studies U.S. households using a detailed Pew/PSRA media survey, then checks robustness with the NLSY and CPS, state-level U.S. data, and cross-country European evidence.
- **Main challenge.** Political activism is not randomly assigned. More educated, wealthier, more sophisticated, or more socially connected people may both vote more and invest more. So a raw correlation between voting and stock ownership is not enough.
- **Main reduced-form result.** In the main household-level regressions, politically active individuals are about **9%** more likely to participate in the stock market, even after controlling for many standard determinants of participation.
- **Main identification result.** Using residence in battleground states and related geographic instruments, the estimated effect becomes even larger: political activism raises participation by roughly **25%** in the IV specifications.
- **Mechanism evidence.** Politically active households spend much more time consuming news, more often seek information about investing, and are more knowledgeable about the economy and markets.
- **Why the paper matters.** The paper adds a new channel to the stock-market-participation literature: **news-related information spillovers** from politics to finance. It is a reduced-form paper about participation costs, not a structural portfolio-choice paper.

2 Data and measurement (key objects)

2.1 Observed PSRA/Pew household survey

- **Main data source.** The main sample is the **PSRA/Pew Poll #2000-MEDIA** survey.
- **Rough sample size.** The core Pew sample contains a little over 3,100 households.
- **Key feature of the survey.** It jointly records:
 - political activism,
 - media choice and time spent on news,
 - direct and indirect stock market participation,
 - demographics, political affiliation, religious affiliation, and state of residence.
- **Political activism variable.** The paper uses the survey question “How often would you say you vote?” and codes a household as politically active if it is registered and reports that it *always* or *nearly always* votes.

- **Participation variable.** The main dependent variable equals one if the household owns stocks either:
 - directly in individual companies, or
 - indirectly through mutual funds that include stock.
- **Direct participation.** In some specifications, the dependent variable is narrower and includes only direct stock ownership.
- **Raw levels.** In the sample, about 59% of households are politically active and about 49% hold stocks directly or indirectly.

2.2 SCF-based imputation variables

- The Pew data do *not* contain household wealth.
- To deal with this, the paper uses the **2001 Survey of Consumer Finances (SCF)** to estimate an imputation model for financial wealth and then predicts wealth in the Pew sample.
- The paper also imputes a **risk-tolerance** measure using the SCF’s self-reported willingness to take financial risk.
- **Why this matters.** Wealth and risk tolerance are first-order controls in any stock market participation regression, so the imputation step is important for identification.

2.3 Other observed data sets and validation samples

- **SCF and HRS.** Used to check whether the Pew sample looks representative in terms of participation rates across demographic groups.
- **NLSY 2008.** Used as a second micro-level robustness check. It contains political activity measures and a broad asset-ownership variable.
- **CPS 2000 Voter Supplement.** Used as another robustness check. CPS lacks household stock ownership, so the authors map households into states with high or low participation using external state-level participation data.
- **State-level U.S. data.** Voter turnout from the U.S. Elections Project and state-level stock market participation from Hong, Kubik, and Stein (2004).
- **Cross-country evidence.** Country-level participation rates from SHARE (for Europe) and HRS (for the U.S.), combined with country-level voter turnout.
- **2006 PSRA/Pew survey.** Used to re-check the mechanism using a later survey wave that contains news-consumption measures but not stock participation.

2.4 Key measured objects

- **Political activism.** Voting regularity and registration.
- **Direct participation.** Ownership of individual-company stock.

- **Total participation.** Direct stock ownership or mutual-fund stock ownership.
- **Media usage.** Time spent reading newspapers, watching TV news, listening to radio news, reading magazines, and, in 2006, getting news online.
- **News-following attitudes.** Whether the respondent enjoys keeping up with the news, and whether the respondent follows different categories of news closely.
- **Knowledge variables.** Knowing who Alan Greenspan is, whether the Fed raised or lowered rates, and the level of the Dow Jones Industrial Average.

2.5 Unobservables and timing

- **Main unobservable.** The paper is really after **information-gathering costs**, which are not directly observed.
- **Other omitted traits.** Trust, sociability, smartness, financial sophistication, and general political preferences could all jointly affect activism and participation.
- **Timing.** The main micro evidence is essentially cross-sectional, so the paper leans heavily on instruments, auxiliary mechanism evidence, and robustness across data sets rather than panel identification.

3 Models and empirical specifications (all equations + interpretation)

3.1 Wealth imputation equation

Because wealth is missing from the Pew survey, the authors estimate a cross-sectional imputation model in the SCF:

$$Wealth_i = \alpha_0 + \alpha_1 Income_i + \alpha_2 Income_i^2 + \alpha_3 Age_i + \alpha_4 Race_i + \alpha_5 Education_i + \alpha_6 Education_i^2 + u_i.$$

- **Interpretation.** This is not part of the economic theory of the paper; it is a data-construction step that gives the authors a proxy for household financial wealth in the main sample.
- **Economic role.** It ensures that the estimated effect of political activism is not simply capturing the effect of wealth.

3.2 Risk-tolerance imputation equation

Similarly, the SCF is used to impute risk tolerance:

$$RiskTol_i = \delta_0 + \delta_1 Income_i + \delta_2 Income_i^2 + \delta_3 Age_i + \delta_4 Age_i^2 + \delta_5 Race_i + \delta_6 Education_i + \delta_7 Education_i^2 + e_i.$$

- **Interpretation.** This creates a control for self-reported willingness to take financial risk.
- **Why it matters.** If politically active people were simply less risk averse, the activism coefficient would not tell us much about information costs. So this control is important conceptually.

3.3 Baseline participation regression

The baseline empirical specification is a probit model:

$$\mathbb{P}(\text{Participation}_i = 1 \mid PA_i, X_i) = \Phi(\beta_0 + \beta_1 PA_i + X_i' \gamma), \quad (1)$$

where:

- *Participation_i* is either total stock market participation or direct participation,
- *PA_i* is the political activism indicator,
- *X_i* includes wealth, income, age, gender, race, education, religion, and imputed risk tolerance.
- **Interpretation.** β_1 is the marginal effect of being politically active on the probability of holding stocks.
- **Main baseline result.** In the richer specifications, political activism raises total participation by about 9.2% and direct participation by about 7.4%.

3.4 Nonlinear and extended baseline specifications

The paper also estimates richer versions of (1), for example:

$$\mathbb{P}(\text{Participation}_i = 1 \mid PA_i, X_i) = \Phi(\beta_0 + \beta_1 PA_i + X_i' \gamma + q(\text{Age}_i, \text{Income}_i, \text{Education}_i)), \quad (2)$$

where $q(\cdot)$ is a collection of squared terms.

It then adds variables like time spent on news and union membership:

$$\mathbb{P}(\text{Participation}_i = 1 \mid PA_i, \text{News}_i, X_i) = \Phi(\beta_0 + \beta_1 PA_i + \beta_2 \text{News}_i + X_i' \gamma). \quad (3)$$

- **Interpretation.** If political activism is partly a proxy for information seeking, then controlling for time spent on news should reduce the coefficient on activism.
- **What happens.** That is exactly what the authors find: once news time is added, the activism effect becomes much weaker, which is consistent with the proposed mechanism.

3.5 State-level augmented regression

To address omitted-variable concerns, the paper adds state-level controls and state fixed effects:

$$\mathbb{P}(\text{Participation}_i = 1 \mid PA_i, X_i, Z_s) = \Phi(\beta_0 + \beta_1 PA_i + X_i' \gamma + Z_s' \lambda + \eta_s), \quad (4)$$

where Z_s includes state-level IQ, investor sophistication, social capital, club meeting attendance, and community-project participation.

- **Interpretation.** This tries to absorb state-level differences in intelligence, sociality, and trust that might otherwise contaminate the activism coefficient.
- **Result.** The political activism effect remains positive and economically large, around 9.2% to 11.6%.

3.6 State-level relation between activism and participation

The paper also studies the simpler state-level relation:

$$Participation_s = \alpha + \rho Turnout_s + W_s' \kappa + \varepsilon_s, \quad (5)$$

where turnout proxies for state-level political activism.

- **Interpretation.** This is not the main identification design, but it is useful because demographic composition varies less across states than across households.
- **Finding.** The state-level economic association between activism and participation is roughly 9.85% to 11.57%.

3.7 Instrumental-variables strategy

The main first stage is:

$$PA_i = \pi_0 + \pi_1 Battleground_s + X_i' \pi + v_i, \quad (6)$$

and the second stage is an IV-probit participation equation:

$$\mathbb{P}(Participation_i = 1 \mid \widehat{PA}_i, X_i) = \Phi(\beta_0 + \beta_1 \widehat{PA}_i + X_i' \gamma). \quad (7)$$

The paper also uses alternative instruments such as:

- perennial battleground states,
- residence in New Hampshire, Iowa, or Washington, DC,
- the voter registration deadline,
- political advertising intensity.
- **Interpretation.** These variables are meant to shift exposure to political news and political engagement, while not directly shifting stock-holding decisions except through information costs.
- **First-stage signs.** Battleground exposure and political advertising raise activism; longer registration deadlines reduce it.
- **Second-stage result.** The IV estimates imply that political activism raises stock market participation by roughly 24% to 39%, with the paper emphasizing an effect of about 25%.

3.8 Cross-country evidence

The paper also studies a country-level relation in Europe:

$$Participation_c = \alpha + \beta PoliticalActivism_c + \varepsilon_c. \quad (8)$$

- **Interpretation.** This is a low-power but useful external check.
- **Finding.** Across 13 European countries, the correlation between political activism and stock market participation is about 0.26.

3.9 Media and knowledge regressions as mechanism evidence

The paper does not build a structural information-acquisition model. Instead, it uses direct mechanism evidence of the form

$$NewsOutcome_i = \alpha + \theta PA_i + X_i' \Gamma + \varepsilon_i, \quad (9)$$

where $NewsOutcome_i$ may be:

- enjoyment of keeping up with the news,
- closeness of following different news categories,
- frequency of gathering investment information,
- minutes spent on news,
- knowledge about Greenspan, the Fed, or the DJIA.
- **Interpretation.** These are mechanism checks: if the channel is lower information cost, politically active households should consume more news and know more about the economy.
- **What the paper finds.** That is exactly what happens.

3.10 Policy counterfactual simulation

The paper also uses a simple “what-if” counterfactual to project how stock market participation would change if other groups had the same activism rate as white males. Abstractly, this is:

$$\widehat{Participation}_g^{cf} \approx Participation_g \times \left(1 + \hat{\beta}^{IV} \cdot \Delta PA_g\right). \quad (10)$$

- **Interpretation.** This is a back-of-the-envelope simulation, not a structural counterfactual.
- **Economic use.** It illustrates how the activism channel may contribute to gaps in participation across demographic groups.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A simple correlation between political activism and stock ownership is not enough.
- Politically active people differ systematically in education, income, age, and possibly trust, social capital, sophistication, or intelligence.
- There is also a joint-determination problem: the decision to vote and the decision to invest may both be outcomes of a deeper underlying trait.

4.2 Why battleground-state exposure helps

- The key idea is that individuals in battleground states are exposed to more political advertising and more intense political coverage.

- This increases political salience and lowers the marginal cost of consuming political news.
- If political and business news are correlated, this creates a spillover into financial information acquisition.
- The identifying assumption is that living in a battleground state affects stock market participation mainly through this political-news channel, not through a direct finance channel.

4.3 Why the additional geographic instruments help

- **Perennial battleground states** capture persistently high political salience.
- **New Hampshire/Iowa/Washington, DC** capture early-primary or permanently political-information-rich locations.
- **Registration deadlines** shift the cost of political participation itself.
- **Political advertising intensity** is a more direct proxy for exposure to campaign information.
- Getting similar estimates across these instruments makes the interpretation more credible.

4.4 Why the mechanism evidence matters for identification

- If activism were only proxying for an omitted preference trait unrelated to information, the paper should *not* systematically find more news consumption and better knowledge among politically active households.
- But it does find exactly that pattern.
- This does not fully prove the exclusion restriction, but it makes the information-cost story much more persuasive.

4.5 Alternative explanations the paper considers

- **Sociability.** The result might reflect social interactions rather than information costs.
- **Trust.** Politically active people may trust institutions more and therefore trust financial markets more.
- **Smartness / sophistication.** Politically active people may simply be smarter or more financially sophisticated.
- **Union membership.** Political activism may just proxy for organized labor exposure.

The paper adds controls or auxiliary evidence for each of these and still finds a positive activism effect.

4.6 Relation to the literature

- It fits into the literature on **limited stock market participation**, especially work emphasizing participation costs and information frictions.

- It complements studies on education, trust, sociability, and IQ as determinants of participation.
- It is also related to the literature showing that politics shapes portfolio choice, but it is distinct because it studies **political activism** rather than ideology or optimism.

5 Tables and figures

5.1 Table 1, Table 2, and Figure 1: Raw data patterns

Table 1
Summary statistics.
This table reports the summary statistics for the key variables used in the empirical analysis. Panel A reports the summary statistics of our main Pw/PSRA data set. Panel B reports the summary statistics for the Survey of Consumer Finances (SCF) data. Panel C compares the stock market participation rates within various demographic groups across the PSRA/Pew, SCF, and Health and Retirement Survey (HRS) data set. We report the participation rates for two groups: (i) Individuals aged 50 or above and (ii) all ages. Panel D reports the summary statistics for the National Longitudinal Survey of the Youth (NLSY) data set and Panel E reports the summary statistics for the Current Population Survey (CPS) data set. All variables are defined in Appendix Table A1.

Variable	Mean	Std dev	Percentile					N
			10th	25th	50th	75th	90th	
Total market participation	0.494	0.500	0	0	0	1	1	3,106
Direct market participation	0.301	0.459	0	0	0	1	1	3,084
Political activism	0.590	0.492	0	0	1	1	1	3,102
Income/\$100,000	0.503	0.387	0.150	0.250	0.350	0.625	0.875	2,568
Age	45.741	18.787	23	31	43	58	74	3,142
Gender	0.476	0.500	0	0	0	1	1	3,142
Race	0.844	0.363	0	1	1	1	1	3,089
Education	12.914	2.500	10	12	13	14	15	3,123
Battleground state (continuous)	0.418	0.493	0	0	0	1	1	2,000
Battleground state (dummy)	0.113	0.086	0.002	0.050	0.100	0.160	0.210	2,000
Religious	0.114	0.318	0	0	0	0	0	3,026
Democrat	0.319	0.466	0	0	0	1	1	3,053
Republican	0.292	0.455	0	0	0	1	1	3,053
Time reading newspapers	12.222	22.544	0	0	0	22,500	45	2,414
Time watching TV	22.574	33.613	0	0	0	45	90	2,243
Time listening to radio	17.993	29.441	0	0	0	22,500	90	3,122
Time reading magazine	13.170	26.681	0	0	0	12,500	45	1,583
Union member	0.139	0.346	0	0	0	0	1	3,084
Perennial battleground	0.161	0.367	0	0	0	0	1	2,006
Registration deadline	22.312	6.637	14	15	25	29	29	1,333
NH	0.024	0.153	0	0	0	0	0	2,006
NH&W	0.048	0.214	0	0	0	0	0	2,000
Washington, DC	0.024	0.153	0	0	0	0	0	2,000
South Carolina	0.015	0.122	0	0	0	0	0	2,000

Panel B: Summary statistics for the SCF sample								
Variable	Mean	Std dev	Percentile					N
			10th	25th	50th	75th	90th	
Financial wealth/\$100,000	1.925	11.267	0.001	0.020	0.222	1.119	3.762	22,210
Risk tolerance	3.125	0.864	3	3	3	4	4	22,210
Income/\$100,000	0.544	0.450	0.15	0.25	0.35	0.625	1.500	22,210
Age	48.959	17.123	28	36	47	61	75	22,210
Race	0.762	0.426	0	1	1	1	1	22,210
Education	13.109	2.917	10	12	13	16	17	22,210

Group	Age > = 50			All ages	
	HRS (%)	Pew (%)	SCF (%)	Pew (%)	SCF (%)
All				49.40	52.34
Age > = 50	33.42	51.77	48.66		
50-65	34.06	57.45	58.71		
> = 65	32.89	43.84	36.95		
Education (in years)					
< 12	11.84	12.42	17.84	13.19	19.82
= 12	32.44	41.76	41.60	38.05	43.77
> 12	33.32	66.42	68.40	61.71	68.47
Gender					
Male	36.02	46.49	54.63	53.17	58.98
Female	31.39	32.54	33.37	43.08	34.24
Income					
Less than \$45,000	29.50	33.31	32.89	33.12	32.89

Table 1 (continued)

Group	Age > = 50			All ages	
	HRS (%)	Pew (%)	SCF (%)	Pew (%)	SCF (%)
\$45,000 or above	51.83	59.38	74.57	63.70	77.93
Race					
White	37.78	53.42	53.21	52.07	57.90
Non-white	12.74	38.46	26.67	38.15	34.53

Variable	Mean	Std dev	Percentiles					N
			10th	25th	50th	75th	90th	
Market participation	0.116	0.320	0	0	0	0	1	7,730
Political activism 1	0.198	0.398	0	0	0	0	1	7,667
Political activism 2	0.736	0.440	0	0	1	1	1	7,118
Strong political activism	0.159	0.366	0	0	0	0	1	7,617
Net Worth/\$100,000	2.621	4.950	0.016	0.199	1.062	2.896	6.105	6,431
Income/\$100,000	0.240	0.391	0.053	0.106	0.173	0.271	0.423	6,086
Age	47.361	2.207	44	45	47	49	51	7,730
Gender	0.488	0.499	0	0	0	1	1	7,730
Race	0.631	0.482	0	0	1	1	1	7,685
Education	0.423	0.494	0	0	0	1	1	7,730
Democrat	0.4687	0.499	0	0	0	1	1	7,247
Republican	0.236	0.424	0	0	0	0	1	7,247
Religion	0.100	0.299	0	0	0	0	0	7,730
Risk tolerance	1.379	0.774	0	1	2	2	2	6,984

Variable	Mean	Std dev	Percentile					N
			10th	25th	50th	75th	90th	
Market participation	0.275	0.076	0.180	0.217	0.285	0.313	0.375	24,831
Political activism	0.060	0.473	0	0	1	1	1	29,676
Income/\$100,000	0.474	0.361	0.112	0.175	0.375	0.675	1.125	24,491
Age	64.475	10.681	52	55	63	73	80	29,676
Gender	0.454	0.497	0	0	0	1	1	29,676
Race	0.883	0.320	0	1	1	1	1	29,676
Education	12.455	2.021	10	12	12	12	15	29,676
Perennial battleground	0.162	0.368	0	0	0	0	1	29,676
Registration deadline	24.622	6.940	15	20	29	30	30	29,676
NH	0.025	0.156	0	0	0	0	0	29,676
NH&W	0.036	0.188	0	0	0	0	0	29,676
Washington, DC	0.011	0.107	0	0	0	0	0	29,676
South Carolina	0.013	0.116	0	0	0	0	0	29,676

Read:

- The raw gap is large. Direct participation is 38.9% for politically active households versus 18.2% for inactive households.
- Total participation is 55.5% versus 32.7%.
- Figure 1 also shows a monotonic relation between how often people vote and how often they hold stocks.
- Table 2 shows that the active–inactive participation gap appears in almost every demographic subgroup, with the main exception being the very lowest-income group.

5.2 Table 3: Baseline probit regressions

Read:

Table 2

Political activism and stock market participation among different cohort groups.

This table reports the level of political activism and the total stock market participation rates, conditional upon household characteristics. Individuals who always or nearly always vote in all elections are identified as politically active. We consider the following household characteristics: Race, gender, political and religious affiliation, education, and income. All variables are defined in Appendix Table A1.

Variable	Politically active (%)	Market participation rate		Difference (%)
		Active (%)	Inactive (%)	
All	59.0	59.7	35.3	24.5
White	60.6	62.3	37.6	24.7
Minorities	51.1	43.5	25.0	18.4
Male	59.8	64.4	36.9	27.5
Female	58.3	55.4	33.8	21.5
Democrat	66.9	53.6	32.5	21.1
Republican	68.7	68.8	44.5	24.3
Protestant	62.6	57.2	32.2	25.1
Catholic	56.8	63.8	39.2	24.6
Years of education				
4	39.8	16.3	6.7	9.6
10	35.0	20.5	14.8	5.7
12	54.6	50.6	30.2	20.4
13	69.7	54.4	34.8	19.6
14	62.9	62.2	48.1	14.1
15	68.9	76.8	60.3	16.5
17	82.5	84.7	72.3	12.4
Income				
0.050	44.6	8.7	9.8	-1.1
0.150	45.0	28.7	14.7	14.1
0.250	45.0	38.2	23.2	15.0
0.350	55.7	53.4	37.2	16.3
0.450	59.6	65.4	50.1	15.4
0.625	67.9	71.2	58.6	12.6
0.875	70.8	83.3	64.5	18.8
1.500	70.0	94.5	76.1	18.3
Age				
18-35	38.0	52.0	33.3	18.7
36-50	62.1	66.4	45.5	20.9
50-65	73.5	67.9	28.3	39.6
65+	80.2	49.1	22.0	27.0

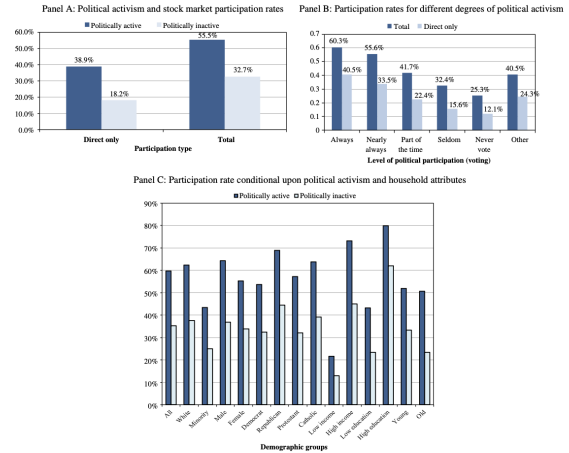


Fig. 1. Political activism and market participation. Panel A reports the stock market participation (direct and indirect), for households by their political activism status. Panel B shows the stock market participation (direct and indirect) for households by their level of political activism. Panel C shows the stock market participation rates for politically active and inactive households, conditional upon various demographic attributes such as race, gender, political and religious affiliations, education, and income. The demographic categories are defined as follows: (i) low income: income/100,000 < 0.25; (ii) high income: income/100,000 > 0.625; (iii) low education: education < = 12 years of school; (iv) high education: education > = 15 years of school; (v) young: age < = 35 years; (vi) old: age > = 65 years. Additional details about the variables are available in Appendix Table A1.

Table 3

Political activism and stock market participation: Baseline estimates.

This table reports the estimation results from probit regressions in which the dependent variable is the stock market participation dummy. In Panel A, the market participation dummy takes a value of one if the household owns stocks directly or indirectly, and zero otherwise. In Panel B, the dummy variable takes a value of one if the household owns stocks directly. In this panel, for brevity, the estimates of control variables are suppressed. The set of independent variables includes political activism, wealth (imputed from the SCF), income, risk tolerance (self-reported), and other demographic variables. The numbers in parentheses below the coefficient estimates indicate the z-statistic for the respective coefficient estimates, which are computed using robust standard errors. Additional details about all variables are available in Appendix Table A1.

Panel A: Dependent variable is total (direct + indirect) participation								
Independent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Political activism	0.241 (13.03)			0.105 (4.40)	0.092 (3.77)	0.091 (3.73)	0.084 (3.43)	0.057 (1.55)
Imputed wealth		0.1752 (16.05)		0.176 (15.52)	-0.101 (-2.54)	-0.011 (-0.39)		-0.051 (-1.19)
Income		0.599 (16.22)			0.370 (3.50)	0.322 (3.01)	1.095 (9.79)	0.116 (2.94)
Age		0.004 (5.61)	-0.011 (-10.26)	-0.011 (-10.73)	0.026 (4.84)	0.020 (4.02)	0.016 (5.14)	0.027 (3.67)
Gender		0.030 (1.37)	0.034 (1.57)	0.029 (1.33)	0.017 (0.76)	0.019 (0.83)	0.019 (0.86)	0.012 (0.37)
Race		0.076 (2.52)	0.017 (0.57)	0.019 (0.65)	-0.024 (-0.68)	-0.073 (-2.15)	0.071 (2.32)	-0.089 (-1.83)
Education		0.063 (11.20)	-0.001 (-0.05)	-0.003 (-0.44)	0.033 (2.34)	-0.008 (-1.04)	0.027 (0.82)	-0.015 (-1.49)
Imputed risk tolerance					0.963 (5.61)	1.082 (6.43)	1.316 (5.41)	1.074 (6.40)
Religious					-0.024 (-0.69)	-0.005 (-0.14)	-0.002 (-0.36)	-0.000 (-0.01)
Squared age								-0.011 (-4.33)
Squared income								-0.361 (-5.12)
Squared education								0.106 (0.81)
Time spent on news							0.001 (1.52)	
Union member								0.007 (0.22)
Number of obs.	3,072	2,428	2,526	2,515	2,515	2,456	2,499	1,119
Pseudo R ²	0.041	0.208	0.198	0.203	0.219	0.217	0.226	0.202
Panel B: Dependent variable is direct participation								
Political activism	0.207 (11.86)			0.079 (3.62)	0.077 (3.56)	0.073 (3.33)	0.073 (3.37)	0.064 (1.98)
Number of obs.	3,050			2,501	2,501	2,442	2,501	1,118
Pseudo R ²	0.037			0.167	0.171	0.170	0.173	0.153

- Once standard controls are added, the effect shrinks from the raw gap but stays economically important.
- The main message is that activism is not just picking up obvious demographic differences.
- The total-participation marginal effect is about 9.2% in the preferred baseline, and the direct-participation effect is about 7.4%.
- When time spent on news is added, the coefficient on activism becomes smaller, which is strong mechanism evidence.

5.3 Figure 2 and Table 4: State-level evidence

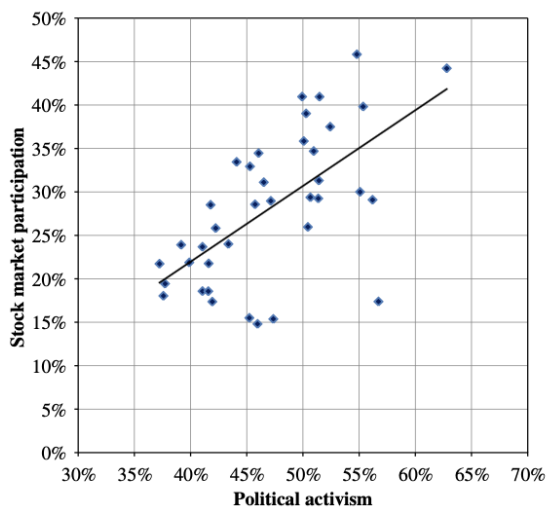


Fig. 2. State-level political activism and stock market participation. This figure plots the relation between state-level political activism (as proxied by the average voter turnout in a state) and the stock market participation rate during the 1980–2008 period. The voter turnout data are from the United States Elections Project (http://elections.gmu.edu/voter_turnout.htm). The state-level stock market participation rates are from Hong, Kubik, and Stein (2004) study. The dots indicate the stock market participation and voter turnout for a given state. The solid line is the fitted line to these dots, which is a linear regression between participation rate and voter turnout. Additional details about the variables are available in Appendix Table A1.

Table 4
Political activism and stock market participation: Estimates using state-level variables.
This table reports the estimation results from extended probit regression specifications in which the dependent variable is the stock market participation dummy. In columns 1, 2, and 3, the dependent variable is the total market participation dummy, while in columns 4, 5, and 6 the dependent variable is the direct participation dummy. The set of independent variables is similar to those used in Table 3. In columns 3 and 6, we include additional control variables, including IQ, investor sophistication, social capital index, and the following two sociability measures: Attended club meetings, worked on community projects. All these variables are imputed at the state-level. The first two variables are taken from the Kornitsis and Kumar (2012) study and the last three variables are from the “Bowling Alone” Web site (<http://www.bowlingalone.com/data.htm>). The numbers in parentheses below the coefficient estimates indicate the z-statistic for the respective coefficient estimates, which are computed using robust standard errors clustered at the state-level. Additional details about all variables are available in Appendix Table A1.

Independent variable	Total participation (1–3)			Direct participation (4–6)		
	(1)	(2)	(3)	(4)	(5)	(6)
Political activism	0.107 (3.41)	0.116 (3.62)	0.101 (3.16)	0.098 (3.81)	0.109 (3.86)	0.092 (3.17)
Imputed wealth	–0.008 (–1.9)	–0.008 (–1.99)	–0.161 (–3.22)	–0.064 (–1.28)	–0.062 (–1.35)	–0.107 (–1.96)
Income	0.314 (2.55)	0.281 (2.16)	0.451 (4.05)	0.388 (3.01)	0.395 (3.23)	0.463 (2.95)
Age	0.028 (3.66)	0.029 (4.24)	0.036 (4.82)	0.015 (2.63)	0.015 (2.49)	0.020 (3.63)
Gender	–0.004 (–0.18)	–0.004 (–0.16)	0.011 (0.55)	0.040 (2.10)	0.039 (1.53)	0.034 (1.82)
Race	0.004 (0.13)	–0.011 (–0.24)	0.024 (0.55)	0.027 (0.68)	0.033 (0.77)	0.032 (0.71)
Education	0.026 (1.62)	0.024 (1.46)	0.043 (2.45)	0.036 (1.75)	0.037 (2.05)	0.046 (1.77)
Imputed risk tolerance	1.055 (4.83)	1.135 (5.08)	1.192 (5.13)	0.422 (2.94)	0.406 (2.31)	0.523 (3.63)
State IQ			0.000 (0.11)			–0.001 (–0.36)
State investor sophistication			0.082 (1.67)			0.075 (1.93)
State social capital index			–0.044 (–0.97)			–0.052 (–1.42)
State: Attended club meetings			0.035 (1.84)			0.042 (2.45)
State: Worked on community projects			–0.016 (–0.49)			–0.040 (–1.62)
State fixed effects	No	Yes	No	No	Yes	No
Number of obs.	1,619	1,618	1,426	1,612	1,599	1,419
Pseudo R ²	0.234	0.252	0.248	0.187	0.206	0.187

Read:

- Figure 2 shows that states with higher voter turnout also have higher stock market participation.
- Table 4 shows that this relation survives adding state fixed effects and state-level controls such as IQ, investor sophistication, and social-capital measures.
- The state-level estimates are around 10% to 12%, close to the household-level baseline results.
- This is important because it weakens the concern that the micro result is driven entirely by household-level omitted traits.

Table 5
Political activism and stock market participation: IV regression estimates.
This table reports the estimation results from instrumental variables (IV) regressions. Panel A reports the results where the dependent variable takes a value of one if the household owns stocks directly or indirectly (total participation), and zero otherwise. Panel B reports the results in which the dependent variable takes a value of one if the household owns stock directly, and zero otherwise. In each panel we report results from the first stage at the top and the IV marginal probability estimates at the bottom. In these two panels, we consider the following instruments: (i) The state battleground status dummy, which takes the value of one if the household lives in a battleground state, (ii) residence in a perennial battleground state, (iii) voting registration deadline, (iv) residence in Iowa, New Hampshire, or Washington, DC, and (v) political advertising (= minutes of political advertising in the state/10,000).
The numbers in parentheses below the coefficient estimates indicate either the t-statistic or the z-statistic for the respective coefficient estimates, which are computed using robust standard errors. All estimates are obtained using a more efficient maximum-likelihood estimator, which does not provide an R² value. Additional details about all variables are available in Appendix Table A1.

Panel A: Dependent variable is total participation					
Independent variable	(1)	(2)	(3)	(4)	(5)
First stage					
Battleground state	0.055 (2.57)				
Perennial battleground		0.034 (2.21)			
Registration deadline			-0.005 (-3.03)		
NHWP				0.166 (3.34)	
Political advertising					0.005 (2.79)
Constant	0.315 (1.44)	0.708 (1.52)	1.194 (2.17)	0.677 (1.46)	0.828 (4.68)
Second stage					
Political activism	0.314 (3.23)	0.328 (2.85)	0.334 (1.88)	0.382 (2.72)	0.312 (2.89)
Number of obs.	1,572	1,557	1,075	1,557	1,477
Log likelihood	-1727.67	-1749.81	-1192.85	-1744.89	-1652.10
Panel B: Dependent variable is direct participation					
Independent variable	(1)	(2)	(3)	(4)	(5)
First stage					
Battleground state	0.046 (2.46)				
Perennial battleground		0.021 (1.73)			
Registration deadline			-0.005 (-3.06)		
NHWP				0.147 (2.87)	
Political advertising					0.005 (2.48)
Constant	0.667 (1.44)	0.643 (1.38)	1.210 (2.20)	0.618 (1.33)	0.676 (1.41)
Second stage					
Political activism	0.244 (2.97)	0.332 (2.37)	0.281 (1.77)	0.394 (3.16)	0.306 (2.18)
Number of obs.	1,565	1,565	1,078	1,565	1,483
Log likelihood	-1699.75	-1701.63	-1158.56	-1697.70	-1613.80

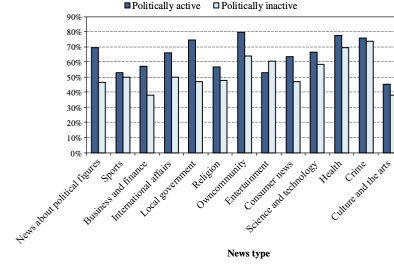


Fig. 3. Political activism and news consumption. This figure shows the fraction of households who closely follow different type of news, separately for politically active and politically inactive households. The survey asks respondents to report about how closely they follow different type of news; specifically, they ask respondents "I'm going to read you a list of different types of news. Please tell me how closely you follow this type of news either in the newspaper, on television, or on radio: very closely, somewhat closely, not very closely, or not at all closely?" Additional details about the variables are available in Appendix Table A1.

5.4 Table 5: IV regressions

Read:

- The battleground-state first stage is positive and significant, as are the alternative news-exposure instruments.
- The IV estimates are much larger than the baseline probit estimates, implying participation effects in the 24% to 39% range.
- The authors interpret the larger IV estimates as evidence that exogenous increases in political information exposure have large participation spillovers.
- This is the paper's core causal table.

5.5 Figure 3, Figure 4, and Table 6: Mechanism evidence on information costs

Read:

- Figure 3 shows that politically active households follow almost every type of news more closely, with the exception of entertainment news.
- Figure 4 shows they also gather investing information more frequently: for example, 5.92% of active households seek investment information every day versus 2.28% of inactive households.
- Table 6 shows a large time-use gap: politically active households spend 72.2 minutes per day on news versus 43.6 minutes for inactive households; in the 2006 survey the corresponding gap is 76.7 versus 48.2.
- Table 6 also shows a knowledge gap: active households are much more likely to know who Alan Greenspan is, whether rates changed, and the level of the DJIA.

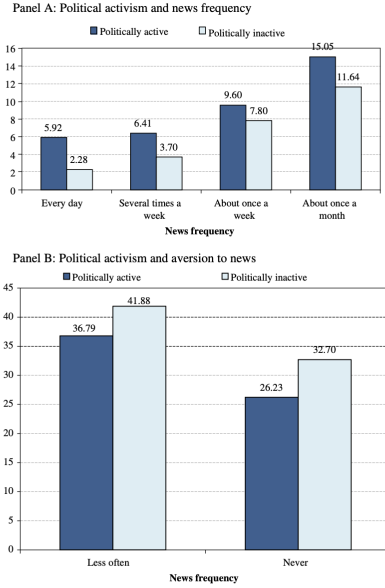


Fig. 4. Political activism and frequency of gathering information. PSRA/Pew asks respondents: "How often, if ever, do you get information from news sources about investing—that is, information that helps you decide how and where to invest your money? Do you do this..." The following answers are provided: (1) every day, (2) several times a week, (3) about once a week, (4) about once a month, (5) less often, (6) never. Panel A shows the fraction of households who choose to gather news with certain frequency, conditional upon the level of political activism. Panel B reports the fraction of households who choose the least frequent answer, conditional upon the level of political activism. Additional details about the variables are available in Appendix Table A1.

Table 6
This table reports the average time spent in minutes on various news sources, conditional upon the level of political activism and the stock market participation status. Respondents report the average amount of time they spent reading newspapers, watching TV, listening to the radio, and reading magazines. Panel B reports the estimates using the 2006 PSRA sample. Panel C reports the fraction of all households/stockholders who report correctly in each category that: (i) Alan Greenspan was the Chairman of Federal Reserve; (ii) the Federal Reserve raised interest rates in recent months; and (iii) the current level of the Dow Jones Industrial Average (DJIA). Additional details about all variables are available in Appendix Table A1.

Panel A: Average time spent gathering information				
News source	Politically active		Politically inactive	
	All	Stockholders	All	Stockholders
Reading newspapers	15.7	16.9	7.0	9.9
Watching TV	27.5	26.3	16.6	17.4
Listening to radio	19.8	22.3	13.5	16.7
Reading magazine	5.1	9.2	6.5	6.8
Total time	72.2	74.7	43.6	50.8

Panel B: Average time spent gathering information, 2006 sample		
News source	Politically active	Politically inactive
Print or online newspaper	18.7	8.3
Watching news on TV	33.4	24.7
Listening to news on the radio	17.7	11.2
Getting news online	7.0	4.1
Total time	76.7	48.2

Panel C: Knowledge of economic environment				
Question	Politically active		Politically inactive	
	All (%)	Stockholders (%)	All (%)	Stockholders (%)
Know Alan Greenspan	56.8	67.5	26.7	47.2
Raised or lowered interest rates	65.5	75.3	37.7	55.1
Level of the DJIA	36.0	48.9	17.7	30.8

5.6 Table 7: Robustness using NLSY and CPS

Read:

- Using the NLSY, alternative measures of political activism still positively predict market participation.
- The implied effects range from about 1.6% to 8.7%, depending on the activism definition.
- Using CPS plus state-level participation assignments, the OLS and IV estimates remain positive, with effects in the 3.7% to 11.5% range.
- So the main finding is not peculiar to the Pew survey alone.

5.7 Appendix Table A2 and Table A4: Cross-country evidence and policy simulation

Read:

- Appendix Table A2 shows that countries with higher political activism tend to have higher stock market participation, though the sample is small.
- Appendix Table A4 gives back-of-the-envelope counterfactuals. If minorities, women, low-income households, and low-education households had the political activism rate of white males, their participation would rise noticeably.
- The largest projected gains are for low-income households, with participation rising by about 4.19 percentage points, and for low-education households, by about 3.22 percentage points.
- These are not structural welfare calculations, but they make the economic relevance of the activism channel concrete.

Table 7

Stock market participation regression estimates using NLSY and CPS data sets.

This table reports the regression estimates from stock market participation regressions using the National Longitudinal Survey of the Youth (NLSY) and Current Population Survey (CPS) data sets. In Panel A, the dependent variable in each regression is the market participation dummy. The key independent variables are the three measures of political activism: (i) Political activism 1, which takes the value of one if the respondent always follows the political news, and zero otherwise; (ii) political activism 2, which takes the value of one if the respondent always votes or voted in the 2006 election year, and zero otherwise; and (iii) strong political activism, which takes the value of one if political activists 1 and 2 dummies are both one, and zero otherwise. Other independent variables are demographic attributes and other investor characteristics that are used in the baseline specification in Table 3. The sample period is 2008. Panel B reports the market participation regression results using the CPS data for the year 2000. In each of these regressions the dependent variable is a binary variable that takes the value of one if the respondent lives in a state with high percentage of stockholders (top quartile), and zero otherwise. The independent variables include the following demographic attributes: income, age, race, gender, education, and marital status, as well as resident seniority (number of years at the current address) and occupation. In addition, we use squared age, squared income, and squared education in the specifications. For brevity, we do not report those estimates. Column 1 reports the estimates using OLS and in the other columns we estimate IV regressions similar to Table 5. Additional details about the instruments are available in Table 5. The numbers in parentheses below the coefficient estimates indicate the z-statistic for the respective estimates, which are computed using robust standard errors. Additional details about all variables are available in Appendix Table A1.

Panel A: Probit regression estimates using the NLSY data								
Independent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Political activism 1	0.036 (3.86)		0.022 (2.35)		0.024 (1.98)		0.016 (1.35)	
Political activism 2		0.105 (16.22)	0.095 (10.99)			0.087 (9.72)	0.071 (5.94)	
Strong political activism				0.069 (6.96)				0.040 (3.27)
Wealth					0.010 (12.43)	0.011 (15.87)	0.009 (11.53)	0.009 (13.06)
Income					0.040 (3.82)	0.043 (5.00)	0.037 (3.35)	0.039 (3.93)
Age					0.002 (0.52)	-0.006 (-4.49)	0.001 (0.54)	0.003 (1.70)
Gender					0.007 (0.71)	0.015 (2.04)	0.013 (1.33)	0.004 (0.46)
Race					0.078 (7.51)	0.081 (10.10)	0.074 (6.98)	0.072 (7.89)
Education					0.015 (1.58)	0.005 (0.65)	0.014 (1.39)	0.020 (2.38)
Risk tolerance					-0.006 (-1.01)	-0.010 (-2.04)	-0.006 (-0.93)	-0.002 (-0.31)
Religious					0.013 (0.78)	0.009 (0.66)	0.015 (0.86)	0.012 (0.81)
Number of obs.	7,667	13,725	7,091	9,276	4,836	8,828	4,517	5,720
Pseudo R ²	0.003	0.028	0.029	0.007	0.080	0.089	0.089	0.083
Panel B: Instrumental variables regression estimates using the CPS data								
Independent variable	Probit	IV (2-6)						
	(1)	(2)	(3)	(4)	(5)	(6)		
<i>First stage</i>								
Battleground state		0.031 (2.92)						
Perennial battleground			0.017 (2.28)					
Registration deadline				-0.002 (-3.56)				
NHW					0.043 (2.77)			
Political advertising							0.046 (3.01)	
Constant		-0.944 (-2.88)	-0.910 (-2.72)	-0.864 (-3.56)	-0.901 (-2.65)		-0.797 (-3.23)	
<i>Coefficient estimates of other variables are suppressed</i>								
<i>Second stage</i>								
Political activism	0.037 (4.87)	0.093 (2.87)	0.090 (2.53)	0.061 (3.26)	0.089 (2.43)	0.115 (2.57)		
Occupation control	Yes	Yes	Yes	Yes	Yes	Yes		
Number of obs.	24,831	24,831	24,831	24,831	24,831	24,831		
Pseudo R ²	0.010							
Log likelihood		-26,572.10	-26,193.34	-26,094.22	-26,117.67	-23,049.83		

Table A2

Political activism and stock market participation: Country-level estimates.

This table reports the stock market participation rate and the level of political activism at the country-level. Column 2 reports the stock market participation rate; column 3 reports the data source; column 4 reports the sample period; and column 5 reports the voter turnout as a proxy for the level of political activism. SHARE is the Survey of Health, Aging and Retirement in Europe. HRS is the Health and Retirement Study administered by the Institute for Social Research at the University of Michigan. Additional details about the variables are available in Appendix Table A1.

Country	Probability of owning stocks or mutual funds	Data set	Years	Political activism
Austria	0.113	SHARE	2004-07	0.716
Germany	0.252	SHARE	2004-07	0.776
Sweden	0.685	SHARE	2004-07	0.819
Netherlands	0.250	SHARE	2004-07	0.800
Spain	0.082	SHARE	2004-07	0.756
Italy	0.110	SHARE	2004-07	0.836
France	0.263	SHARE	2004-07	0.839
Denmark	0.466	SHARE	2004-07	0.855
Greece	0.058	SHARE	2004-07	0.753
Switzerland	0.386	SHARE	2004-07	0.482
Belgium	0.289	SHARE	2004-07	0.910
Czech Republic	0.061	SHARE	2006-07	0.644
Poland	0.019	SHARE	2006-07	0.509
USA	0.268	HRS	1992-2008	0.560

Table A4

Impact of increase in political activism on participation decisions: Simulation results.

This table reports simulation results from "what-if" scenarios in which we predict the direct and indirect stock market participation rates for the same level of political activism as white males. Columns 2 and 6 labeled as "Data" report the actual political activism levels and stock market participation rates for White males, Minorities, and women, respectively. Columns 3 and 7 labeled as "Projection" report the political activism levels and stock market participation rates for minorities and women, should they have the same level of political activism as White males. Columns 4 labeled as "% Increase" report the percent increase in the stock market participation rate, which is computed as 100*(projection/data)-1. Finally, columns 5 and 9 labeled as "In percentage" report the percentage rate increase in the stock market participation rates, which is computed as 100*(projection-data). Additional details about the variables are available in Appendix Table A1.

Variable	White male					Minorities (2-5)				Women (6-9)			
	Baseline (1)	Data (2)	Projection (3)	% Increase (4)	In percentage (5)	Data (6)	Projection (7)	% Increase (8)	In percentage (9)	Data (6)	Projection (7)	% Increase (8)	In percentage (9)
Political activism	0.616	0.511	0.615	20.40	10.45	0.583	0.616	5.70	3.34	0.583	0.616	5.70	3.34
Participation	0.562	0.351	0.377	7.40	2.61	0.459	0.468	1.80	0.83	0.459	0.468	1.80	0.83
Variable	White male					Low income (2-5)				Low education (6-9)			
	Baseline (1)	Data (2)	Projection (3)	% Increase (4)	In percentage (5)	Data (6)	Projection (7)	% Increase (8)	In percentage (9)	Data (6)	Projection (7)	% Increase (8)	In percentage (9)
Political activism	0.616	0.448	0.616	37.30	16.74	0.487	0.616	26.50	12.90	0.487	0.616	26.50	12.90
Participation	0.562	0.166	0.208	25.20	4.19	0.328	0.360	9.80	3.22	0.328	0.360	9.80	3.22

6 Short summary

- **Main conclusion.** Politically active households are more likely to hold stocks.
- **Mechanism.** The evidence points to lower information-gathering costs: politically active people enjoy news more, consume more news, and know more about the economy.
- **Identification.** Geographic variation in exposure to politics—especially battleground-state residence—suggests a causal effect, not just a correlation.
- **Quantitative lesson.** The effect is economically meaningful: about 9% in baseline regressions and around 25% in IV.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that political activism is an important predictor of stock market participation.
- More importantly, it argues that the link is not merely descriptive: it reflects an information-cost channel.
- The identification strategy is not perfect, but taken together with the mechanism evidence and robustness checks, it gives a persuasive case that greater political engagement lowers the cost of becoming informed about financial markets.

7.2 Economic intuition

- Many households do not hold stocks because participating requires effort: gathering information, interpreting it, and feeling informed enough to invest.
- Politically active people already pay part of this fixed cost for another reason—they like to keep up with the news and political process.
- Once that fixed cost is partly paid, the marginal cost of learning about markets falls, so participation becomes more attractive.
- In the paper’s language, politics creates a positive information spillover into finance.

7.3 Takeaway

This paper is a reduced-form contribution to the stock-market-participation literature. It does not build a full structural model of household portfolio choice. Instead, it isolates a novel and intuitive channel: households that are more engaged in politics behave like lower-information-cost investors. The evidence is especially compelling because the paper combines household regressions, geographic instruments, mechanism tests on news consumption and economic knowledge, state-level evidence, and robustness across multiple data sets. For work on participation costs, financial literacy, media exposure, or the political determinants of household finance, this paper is a useful benchmark.